Blessed G/Michael Catholic Secondary School
Second Semester Physics Test one for Grade 10th

March, 2018E.C **Time Allowed 1: 00 hrs**

Name Bitanya Wedinagegnehu **Section** 10th B **No** 12

I. *Write 'True' if the statement is correct and 'False' if it is incorrect*

True 1. In a series circuit, current is the same through all components.
True 2. Magnetic force is maximum when velocity of charge is perpendicular to the field.
false 3. A moving charge experiences a magnetic force if it moves parallel to the field.
false 4. Magnetic field is produced only by magnets, not by currents.
false 5. Magnetic field lines can intersect each other.
false 6. Electric field lines can originate from negative charges.
True 7. Increasing the cross-sectional area of a wire decreases its resistance. $R \propto \frac{1}{a}$
True 8. In parallel combination of resistance the equivalent resistance is smaller than the smallest of all.
false 9. Conductivity and resistivity has direct relation with each other
false 10. Magnetic field lines pointing from the South to the North Pole outside a bar magnet.

II. *Choose the best answer for the following questions.*

BD 11. A wire of resistance R is stretched to double its length. What is the new resistance?
A) R B) 2R C) 3R D) 4R

AD 12. The SI unit of electric field is:
A) N/C B) J/C C) V/m D) Both A and C

C 13. If two resistors (2Ω and 4Ω) are connected in parallel, the equivalent resistance is:
A) 6Ω B) 3Ω C) 1.33Ω D) 2Ω

D 14. Resistance of a conductor depends on:
A) Length B) Area C) Material D) All of the above

B 15. Magnetic field around a current-carrying wire is:
A) Straight B) Circular C) Random D) Zero

A 16. Three resistors 2Ω, 3Ω, and 6Ω are in parallel. Equivalent resistance is:
A) 1Ω B) 2Ω C) 3Ω D) 6Ω

A 17. An electron moves with velocity 10^6 m/s perpendicular to a 0.1 T magnetic field. Force is $e = 1.6 \times$
A) 1.6×10^{-14} N B) 1.6×10^{-15} N C) 1.6×10^{-13} N D) 1.6×10^{-16} N

A 18. Magnetic Force on Moving Charge $F = qvB\sin\theta$. Maximum force occurs when θ equal to
A) 0° B) 45° C) 90° D) 180°

C 19. Which statement is correct?
A) Magnetic field lines start from north pole and end at south pole outside the magnet
B) Magnetic field lines intersect each other
C) Magnetic field lines are always straight
D) Magnetic field lines can exist in a vacuum only

B 20. A straight conductor of length 0.5m carries a current of 4A and is placed perpendicular to a uniform magnetic field of 0.3T. What is the magnetic force on the conductor?
A) 0.3 N B) 0.6 N C) 2 N D) 6 N

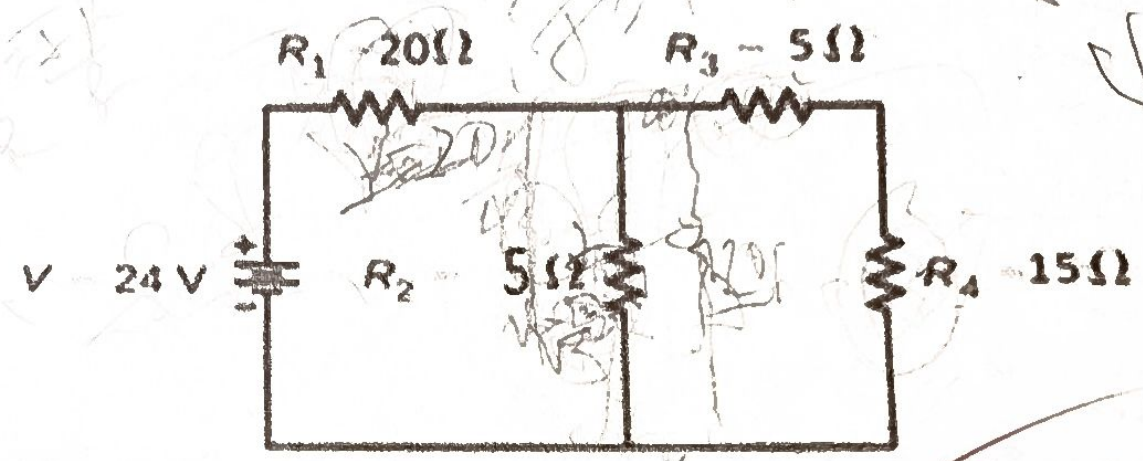
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A 21. Two long parallel currents attract each other if:
A) Currents in same direction C) One current zero
B) Currents in opposite direction D) None

B 22. A stationary charge is placed in a uniform magnetic field. Force on it is:
A) Maximum C) Depends on charge magnitude
B) Zero D) Depends on field direction

D 23. Which one of the following statement is true?
A) Magnetic forces are weak at the poles of a magnet.
B) Magnetic field lines can cross each other.
C) The Earth's magnetic field is strongest at the geographic poles.
D) Denser lines indicate a stronger field.

III. **<u>Use the figure shown below to solve the question listed below</u>**



23. Calculate the equivalent resistance $R_{equivalent}$ = 24Ω

24. Calculate current in each resistor

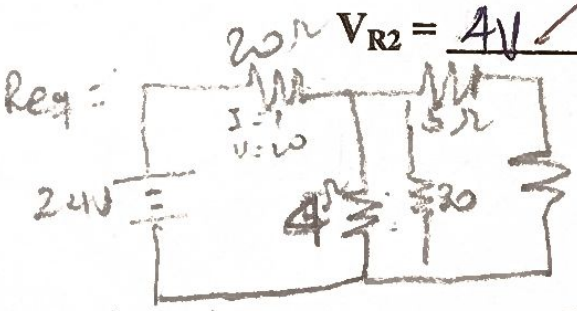
I_{R1} = 1A I_{R3} = 0.8A

I_{R2} = 0.8A I_{R4} = 0.26A

25. Calculate the voltage in each resistor

V_{R1} = 20V V_{R3} = 4V

V_{R2} = 4V V_{R4} = 4V

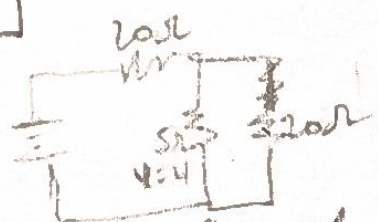


$\frac{1}{5}+\frac{1}{20}=\left(\frac{4+1}{20}\right)^{-1}=4$

$I_T=\frac{V}{R}=\frac{24}{24}$

$V=20\times1=20=V_1$

$V_2=24-20=4V$



$I_2=\frac{V}{R}=\frac{4}{5}=0.8A$

V: 4V

$I_3=0.2$ but $I_3+I_2=1$

$V=I.R=0.2\times20=4$



Prepared by:- physics department